Ericsson PE (JAY-PE-09) QoS Congestion Event SOP (ALARM_3147)

Node: JAY-PE-09 | **Vendor:** Ericsson | **Severity:** Warning | **Alarm Code:** ALARM_3147

Description

This SOP addresses the QoS Congestion Event (Microbursts, interface tail drops, QoS queue starvation) failure domain on the Ericsson PE (JAY-PE-09) at the Jayanagar Metro Site. The underlying protocol mechanics involve the PE's Quality of Service (QoS) architecture, which is responsible for classifying and prioritizing traffic flows across the network. The PE's QoS framework relies on the Internet Protocol (IP) Differentiated Services Code Point (DSCP) and the Resource Reservation Protocol (RSVP) to enforce traffic contracts. However, in the presence of microbursts or high-priority traffic, the PE's QoS queue may become oversubscribed, leading to interface tail drops and subsequent congestion events. This SOP aims to diagnose and resolve such events by leveraging Ericsson's CLI dialect to investigate the root causes of the congestion event.

Pre-requisite Checks

- Verify that the Jayanagar Metro Site is accessible and that all necessary access permissions are granted.

- Check the redundant link status to ensure that it is operational and functioning correctly.

- Gather relevant system logs and configuration files for further analysis.

Potential Root Causes

Cause	Probability
Incorrectly configured QoS policies or misaligned traffic contracts.	High
Microbursts or high-priority traffic overwhelming the PE's QoS queue.	Medium
Hardware or software issues affecting the PE's interface or QoS processing capabilities.	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy all, Expected Output: Display all QoS policies configured on the PE., Description: This command allows us to inspect the PE's QoS policy configuration, including the classification criteria, queueing strategies, and policing rules.

Command: show interface all, Expected Output: Display the status of all interfaces on the PE., Description: This command enables us to examine the interface configuration, including the queue depths, buffer sizes, and error counters.

Command: show qos queue all, Expected Output: Display the status of all QoS queues on the PE., Description: This command allows us to investigate the QoS queue configuration, including the queue lengths,
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packet drops, and transmission rates.
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Resolution Steps

Step 1: Step: Verify QoS policy configuration and adjust as necessary., Command: configure replace qoS policy , Description: This step involves reviewing the QoS policy configuration and making adjustments as required to ensure that traffic contracts are properly enforced and QoS queues are not overwhelmed.

Step 2: Step: Adjust interface configuration to mitigate microbursts., Command: configure replace interface queue-depth , Description: This step involves modifying the interface configuration to increase the queue depth and mitigate the impact of microbursts on the QoS queue.

Step 3: Step: Perform software or hardware maintenance as necessary., Command: maintenance perform , Description: This step involves performing software or hardware maintenance as required to address any underlying issues affecting the PE's interface or QoS processing capabilities.

Post-Check Verification

```
Command: show qos policy all, Expected Output: Display all QoS
policies configured on the PE., Description: This command allows us to
verify that the QoS policy configuration has been updated correctly
and that traffic contracts are being properly enforced.

Command: show interface all, Expected Output: Display the status of
all interfaces on the PE., Description: This command enables us to
verify that the interface configuration has been updated correctly and
that the queue depth has been increased to mitigate microbursts.

Command: show qos queue all, Expected Output: Display the status of
all QoS queues on the PE., Description: This command allows us to
verify that the QoS queue configuration has been updated correctly and
that packet drops have been mitigated.
```

Roll-back Procedures

- Step: Restore original QoS policy configuration., Command: configure replace qoS policy restore, Description: This step involves restoring the original QoS policy configuration in case the fix fails.

- Step: Revert interface configuration changes., Command: configure replace interface queue-depth , Description: This step involves reverting the interface configuration changes in case the fix fails.

- Step: Revert software or hardware maintenance changes., Command: maintenance revert , Description: This step involves reverting the software or hardware maintenance changes in case the fix fails.